INL1, Problem 6
MAA155

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Exercise 4.20 Let M be a set.

(ii) Let r by a relation on M that is irreflexive, symmetric and transitive.

Indirect Proof: Suppose that $r \neq \phi$

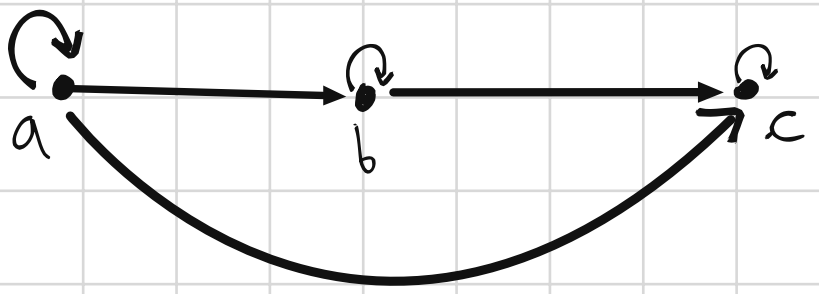
So let $a, b \in M$ with $a\,r\,b$, by symmetry of r, we also have $b\,r\,a$

by transitivity of r, we have $a\,r\,b$ and $b\,r\,a \Rightarrow a\,r\,a$ ↯ Contradiction since r is irreflexive ($\forall a \in M,\ a \not r a$)

Exercise 4.21

(i) example of a relation that is reflexive, transitive but not symmetric

$$r = \{(a,a),(b,b),(c,c),(a,b),(b,c),(a,c)\}$$



Exercise 4.50

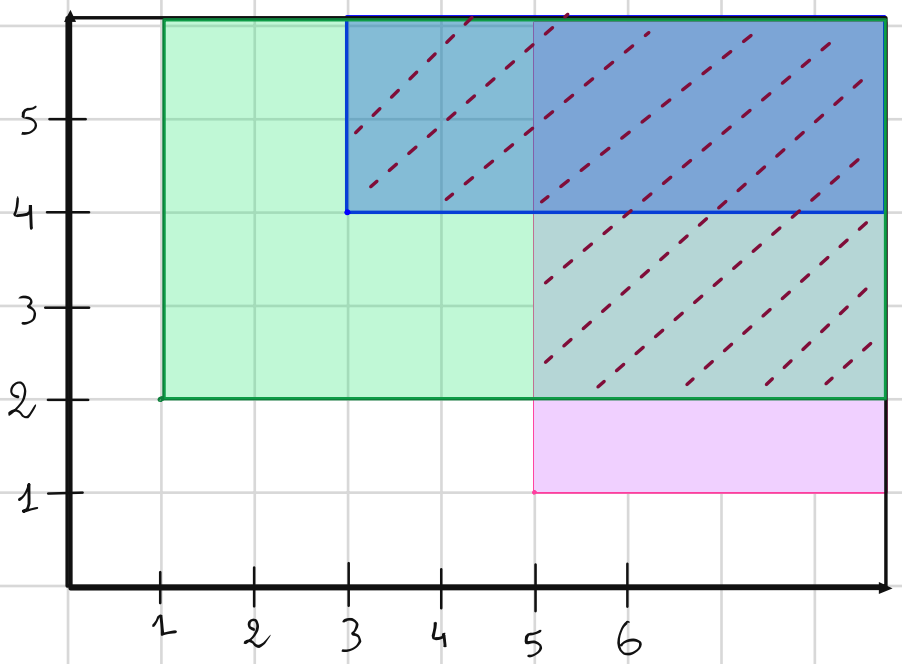
$$A = \{(m,n) \in \mathbb{N}^2 / 3 \leq m, 4 \leq n\}$$

$$B = \{(m,n) \in \mathbb{N}^2 / 5 \leq m, 1 \leq n\}$$

$$C = \{(m,n) \in \mathbb{N}^2 / 1 \leq m, 2 \leq n\}$$

$$D = (A \cup B) \cap C$$

basis of $D \{(3,4),(5,2)\}$



Exercise 4.61:

(i) we have $(0_1,\dots,0_m) \leqslant (a_1,\dots,a_m)$ since either $\forall i \in [\![1,m]\!], a_i = 0$ therefore

$(0_1,\dots,0_m) = (a_1,\dots,a_m)$, or $\exists i \in [\![1,m]\!], a_j = 0, \forall j \in [\![1,i-1]\!]$ and $a_i > 0$

☆ let $(a_1,a_2,\dots,a_m) < (b_1,\dots,b_m)$ and $c = (c_1,\dots,c_m)$

so $a+c = (a_1+c_1, a_2+c_2, \dots, a_m+c_m)$ and $b+c = (b_1+c_1, b_2+c_2, \dots, b_m+c_m)$

$\exists i \in [\![1,m]\!], b_j = a_j, \forall j \in [\![1,i-1]\!]$ and $a_i < b_i$

then $b_j + c_j = a_j + c_j, \forall j \in [\![1,i-1]\!]$ and $a_i + c_i < b_i + c_i$ so $(a+c) \leqslant (b+c)$

(ii) we have $(0_1,\dots,0_m) \leqslant (a_1,\dots,a_m)$ since either $\forall i \in [\![1,m]\!], a_i = 0$ therefore

$(0_1,\dots,0_m) = (a_1,\dots,a_m)$, or $\exists i \in [\![1,m]\!], a_j = 0, \forall j \in [\![i+1,m]\!]$ and $a_i > 0$

☆ let $(a_1,a_2,\dots,a_m) < (b_1,\dots,b_m)$ and $c = (c_1,\dots,c_m)$

so $a+c = (a_1+c_1, a_2+c_2, \dots, a_m+c_m)$ and $b+c = (b_1+c_1, b_2+c_2, \dots, b_m+c_m)$

$\exists i \in [\![1,m]\!], b_j = a_j, \forall j \in [\![i+1,m]\!]$ and $a_i < b_i$

then $b_j + c_j = a_j + c_j, \forall j \in [\![i+1,m]\!]$ and $a_i + c_i < b_i + c_i$ so $(a+c) \leqslant (b+c)$

(iii) we have $(0_1,\dots,0_m) \leqslant (a_1,\dots,a_m)$ since either $\forall i \in [\![1,m]\!], a_i = 0$ therefore

$(0_1,\dots,0_m) = (a_1,\dots,a_m)$, or $\exists i \in [\![1,m]\!]$, $a_j = 0, \forall j \in [\![i+1,m]\!]$ and $a_i > 0$, DO $0 < \sum_{i=1}^{m} a_i$

★ let $(a_1,a_2,\dots,a_m) < (b_1,\dots,b_m)$ and $c = (c_1,\dots,c_m)$

DO $a+c = (a_1+c_1, a_2+c_2, \dots, a_m+c_m)$ and $b+c = (b_1+c_1, b_2+c_2, \dots, b_m+c_m)$

$$\sum_{i=1}^{m} a_i < \sum_{i=1}^{m} b_i \text{ DO } \sum_{i=1}^{m} a_i + \sum_{i=1}^{m} c_i < \sum_{i=1}^{m} b_i + \sum_{i=1}^{m} c_i \text{ DO } (a+c) \leqslant (b+c)$$

(iV) we have $(0_1,\dots,0_m) \leqslant (a_1,\dots,a_m)$ since either $\forall i \in [\![1,m]\!], a_i = 0$ therefore

$(0_1,\dots,0_m) = (a_1,\dots,a_m)$, or $\exists i \in [\![1,m]\!]$, $a_j = 0, \forall j \in [\![i+1,m]\!]$ and $a_i > 0$ DO $\begin{cases} (0_{i+1},\dots,0_m) \leqslant_2 (a_{i+1},\dots,a_m) \\ (0_1,\dots,0_{i-1}) = (a_1,\dots,a_{i-1}) \end{cases}$

★ let $(a_1,a_2,\dots,a_m) < (b_1,\dots,b_m)$ and $c = (c_1,\dots,c_m)$

DO $a+c = (a_1+c_1, a_2+c_2, \dots, a_m+c_m)$ and $b+c = (b_1+c_1, b_2+c_2, \dots, b_m+c_m)$

First case, if $(a_1,\dots,a_i) \leqslant_1 (b_1,\dots,b_i)$

we then have $(a_1+c_1,\dots,a_i+c_i) \leqslant_1 (b_1+c_1,\dots,b_i+c_i)$ because $\leqslant_1$ is admissible

Second Case, we have $(a_1,\dots,a_i) = (b_1,\dots,b_i)$ and $(a_{i+1},\dots,a_m) \leqslant_2 (b_{i+1},\dots,b_m)$

then $(a_1+c_1,\dots,a_i+c_i) = (b_1+c_1,\dots,b_i+c_i)$ and

$(a_{i+1}+c_{i+1},\dots,a_m+c_m) \leqslant_1 (b_{i+1}+c_{i+1},\dots,b_m+c_m)$ because $\leqslant_2$ is admissible.